Earth and Planetary Science Letters, 26 (1975) 263–268
© Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

REMANENT MAGNETISM OF THE TWIN SISTERS DUNITE INTRUSION AND IMPLICATIONS FOR THE TECTONICS OF THE WESTERN CORDILLERA

MYRL E. BECK, Jr.

Department of Geology, Western Washington State College, Bellingham, Wash. (USA)

Received February 5, 1975
Revised version received April 4, 1975

The Twin Sisters Dunite Intrusion has a homogeneous remanent magnetization directed nearly due east and inclined about 60° below the horizontal. It was emplaced in the Early Tertiary as a solid intrusion, but probably was above its Curie temperature at the time and so became magnetized substantially in situ. If so, it must have undergone considerable tectonic disturbance subsequent to emplacement and magnetization, including an important component of clockwise rotation. Examination of paleomagnetic data for the westernmost Cordillera shows that an abnormal number of rock bodies have discordant directions of remanent magnetization, mostly clockwise rotations of declination, flattened inclinations, or both. This sense of discordance in turn suggests that the western margin of the North American plate may have been deformed intermittently by right-lateral shear for at least some few tens of millions of years.

1. Introduction

The Twin Sisters Intrusion of northwestern Washington is well known to petrologists and geochemists as a source of nearly pure, unaltered dunite for laboratory experiments, where it is often cast in the role of a sample of the earth's mantle. According to Ragan [1], it consists essentially of four minerals, olivine (Fo_{90} or more), enstatite, chromite, and clinopyroxene, with olivine greatly predominant. Unlike most Alpine ultramafic bodies, the Twin Sisters Intrusion is almost completely unserpentinized except at its margin [1], although a model calculated by Thompson [2] from gravity and aeromagnetic data indicates the presence of a greater amount of serpentine at depth. Gravity and magnetic measurements also show that the intrusion is relatively thin and lens shaped, with a maximum vertical dimension of a few kilometers [2,3]. Internally the intrusion exhibits several superimposed stages of solid-state deformation, characterized by crystallographic orientation of olivine grains, folding and displacement of mineral bands, cleavage, cataclasis, and the like. Ragan [1], who describes these structures in detail, believes that they developed at temperatures of 500°C or a few hundred degrees higher. The intrusion is believed by most investigators to have been emplaced in the solid-state at a temperature well below the solidus. However, Ragan's estimates of temperature and the paleomagnetic evidence cited below suggest that it was above its Curie temperature when emplaced and thus acquired its magnetization substantially it situ.

Fig. 1 (insert) is a simplified version of a geological map of the northern Cascade Range by Misch [4], showing the complicated regional setting of the Twin Sisters Intrusion. Spatial and temporal association of the intrusion with the Shuksan thrust (Fig. 1) have led most investigators to conclude that emplacement of the dunite mass in the upper crust was controlled by this major structural feature. The Shuksan thrust has itself been folded [3,4], and as a result the Twin Sisters Intrusion has probably been tilted some few tens of degrees to the W or WSW since arriving at high levels in the crust.

Systematic paleomagnetic measurements on the Twin Sisters Intrusion were begun in 1970 by Kinoshita and Beck [5], who also studied the much smaller Goat Mountain Intrusion located a few kilometers

south of the Twin Sisters body. Sampling in the Twin Sisters was confined to the area called the Cirque Section in the present report, located at the northern end of the body (Fig. 1). Unlike the Twin Sisters, the Goat Mountain ultramafic body is strongly serpentinized. Results of this preliminary investigation showed that (1) mean directions of magnetization at the two localities diverge markedly from one another, and (2) neither match "expected" direction of magnetization based on standard polar wandering curves for North America (e.g. [6,7]). The Goat Mountain body is small and intensely altered and so its divergent direction of magnetization was not unexpected. The Twin Sisters body however required further investigation, first to see whether or not it was homogeneously magnetized throughout, and second, if possible to provide an explanation for its strongly divergent magnetic direction. Thus the present investigation was initiated.

2. Methods of investigation

Owing to the rugged, partially glacier-covered and wholly roadless nature of the terrain it was necessary to confine systematic sampling to a few areas surrounding helicopter-supplied base camps. At each base camp from 6 to 9 sampling sites were occupied at intervals of a few hundred meters to about a half-kilometer about the base station. All sites were natural exposures subject to some weathering, but the nearly continuous nature of the rock exposure made it possible to choose relatively little-weathered and massive material to sample. Cores up to 12 cm in length were drilled and oriented in situ by standard paleomagnetic techniques. Each site consisted of from 4 to 9 individually oriented samples, separated from one another by distances of a few meters to a few tens of meters and wherever possible drilled from separate dunite blocks. Samples from the Cirque Section were collected as oriented hand samples over a period of years by H. Kinoshita, S. Beske-Diehl, J. Diehl, J. Whitney and the author. The four intensely sampled sections plus the location of several oriented hand samples collected elsewhere in the range are shown in Fig. 1.

In the laboratory all samples were cut into cylindrical specimens approximately 2.3 cm long and natural remanent magnetization (NRM) for at least one specimen per sample measured on a Schonstedt spinner magnetometer. Initial intensities of remanent magnetization for unserpentinized samples for the most part fell in the 10^{-4} cgs range. Initial scatter was large for most sites but was greatly reduced by alternating-field (a.f.) demagnetization in peak fields of 150–250 Oe, using an apparatus employing a 4-axis Lillard-type tumbler, a General Electric Inductorol for smooth lowering of the field, and a two-axis Helmholz Coil system to cancel most of the ambient geomagnetic field. The optimum field for magnetic cleaning was selected on the basis of minimum dispersion. Most samples tended to be moderately "soft" magnetically; after a.f. demagnetization at 200 Oe the median sample had an intensity only about 12% of its initial value. However, acceptable groupings were obtained for most sites, and as these groupings themselves tended to cluster at a high angle to the present geomagnetic field direction it is apparent that the Twin Sisters dunite retains a component of stable primary remanent magnetization.

To improve the definition of the mean primary magnetic direction for the body, somewhat arbitrary and partially subjective rejection criteria were adopted. First, within-site precision was improved by dropping samples that remained markedly outside the cluster after demagnetization. Such samples were rarely less than 40° from the mean of the group, and in all cases diverged from the mean by at least twice the angular standard deviation [6]. Second, entire sites were dropped if their 95% confidence circles exceeded 20° after individual aberrant samples were rejected. It was evident from examination of the demagnetization curves that these procedures resulted in the rejection of the magnetically less hard and less stable samples. Nevertheless, they cannot presently be defended on rock-magnetics grounds because the mineralogical and petrological characteristics that determine relative stability in these rocks have not been investigated.

Results of laboratory measurements on the Twin Sisters Intrusion are summarized in Table 1 and mean directions for stable sites shown in Fig. 2. For the body as a whole the mean direction (based on 17 sites, the entire Cirque Section being treated as a single site) is: D, 92.5°; I, 59.5°; with 95% confidence interval 5.4° and precision parameter 43.95. The paleomagnetic pole corresponding to this direction is 27.5°N, 62.5°W,

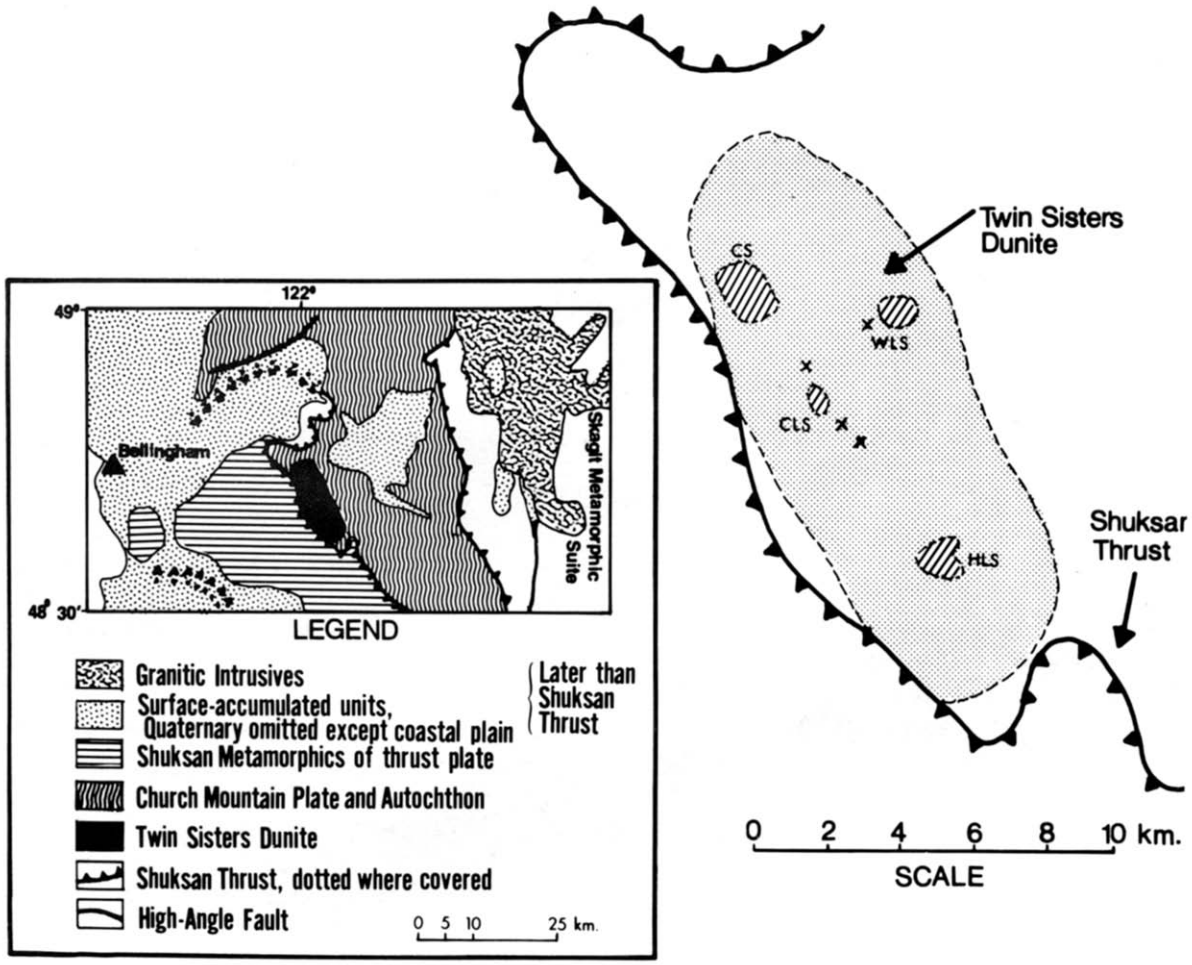


Fig. 1. Geological map, modified from Misch [4], and location map showing sampled sections. CS = Cirque Section; WLS = Wiseman Lake Section; CLS = Cinderella Lake Section; HLS = Heart Lake Section. Small X's show location of scattered hand samples not included in the statistical summary of Table 1.

with an oval of 95% confidence δ_m, 8.2° δ_p 6.1°. All stable samples are of normal (inclination downward) polarity.

3. Discussion

Fig. 2 shows that there is very little difference between the mean directions of remanent magnetization obtained after a.f. demagnetization at the four sampling sections; what differences exist are not statistically significant. As the four sections are scattered widely throughout the body, it seems reasonable to conclude that the entire intrusion shares this magnetic direction. This conclusion is supported by results from scattered samples (Fig. 1) collected elsewhere in the range. In view of the thin, lens-like shape of the body and the abundant evidence of its internal deformation, it appears unlikely that a homogeneous magnetization acquired before emplacement could have survived that process unscathed. Thus the magnetic evidence supports the temperature estimates of Ragan [1] and suggests that the magnetization of the Twin Sisters Intrusion is related to the later stages of its (solid-state) injection in the upper crust. The magnetic direction shown in Fig. 2 can in that case

TABLE 1

Site-mean directions of magnetization, Twin Sisters Intrusion

Site No.	N	D	I	k	α_{95}	Samples omitted
Heart Lake Section (48.65°N, 121. 92°W)						
H50	6	121.5°	67.5°	47.9	9.8°	1
H52	7	68.0	50.0	11.3	19.0	
H53	6	99.5	61.0	16.2	17.0	
H54	3	74.5	63.5	52.2	17.0	2
H55	8	110.5	39.0	24.5	11.5	1
H57	4	101.0	61.5	50.1	13.0	2
H58	9	100.0	47.0	16.1	13.0	
Mean, HLS	(7)	96.0	57.0	30.9	11.0	(2)
Wiseman Lake Section (48.71°N, 121.95°W)						
W59	7	60	77.5	27.5	13.0	1
W62	7	99.5	53.0	26.6	13.0	
W63	5	109.5	57.0	44.5	11.5	2
Mean, WLS	(4)	98.5	61.6	35.3	15.5	(3)
Cinderella Lake Section (48.68°N, 121.98°W)						
C66	5	82.0	54.5	20.5	17.5	1
C68	5	84.0	62.5	48.6	11.0	1
C69	4	61.0	67.0	22.9	19.5	
C70	7	81.0	62.5	28.2	11.5	
C71	6	89.0	62.0	70.4	8.0	
Mean, CLS	(5)	80.0	62.0	154.6	6.0	(1)
Cirque Section (48.72°N, 121.99°W)						
	11	98.0	53.0	30.1	8.5	2

Entries in parentheses under "N" and "Samples omitted" columns refer to number of sites.

be compared with directions calculated from the polar wandering curve of North America. It is not, in all probability, an artifact of magnetization acquired in the mantle at some unknown stage in the distant geological past.

In addition to magnetic directions for the Twin Sisters Intrusion, Fig. 2 shows the present geomagnetic field direction at the sampling site and the approximate direction of the Early Tertiary field at the same locality. The Early Tertiary direction was calculated from Irving and Park's [8] mean North American Lower Tertiary pole, recalculated free of poles from the western edge of the continent, which, as discussed below, may be subject to large-scale tectonic disturbances. The "expected" direction based on this

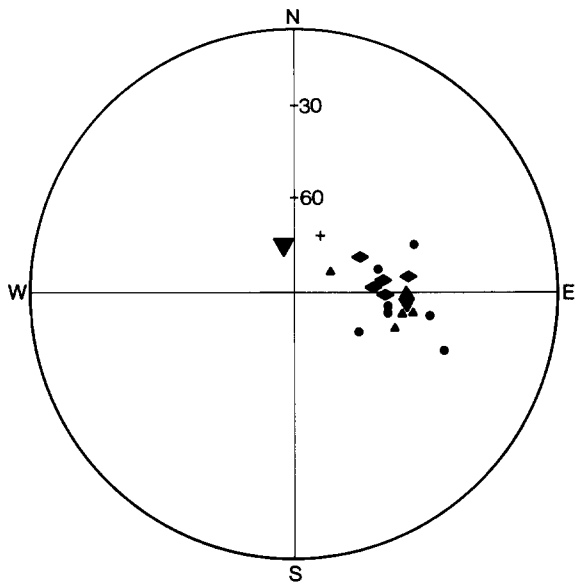


Fig. 2. Equal-area projection of site-mean directions of magnetization for the Twin Sisters Intrusion. Circles: Heart Lake Section; triangles: Wiseman Lake Section; diamonds: Cinderalla Lake Section; large upright diamond: Cirque Section. Direction of the present geomagnetic field shown by a plus symbol; the large inverted triangle shows the direction of the Early Tertiary geomagnetic field at the sampling site (see text).

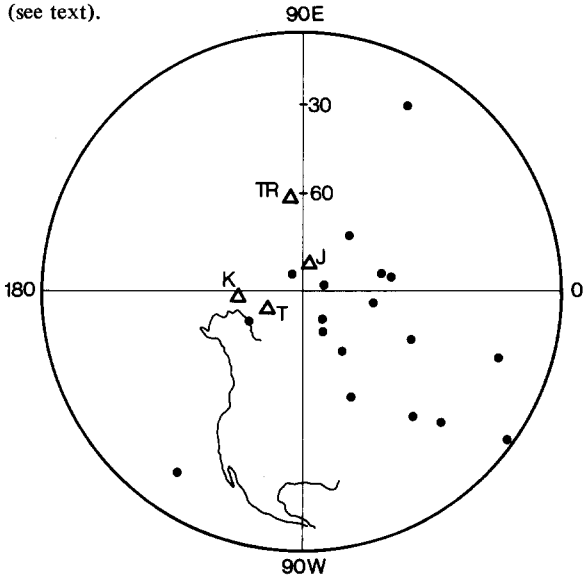


Fig. 3. Polar equal-area plot of discordant pole positions from the westernmost Cordillera of North America. Filled circles are the discordant poles. Triangles show mean poles for North America, calculated free of any poles from the western Cordillera; T = Early Tertiary; K = Cretaceous; J = Jurassic; Tr = Triassic. For a more complete discussion of this diagram, see reference [9].

recalculated pole is 347.5°, 74° (declination, inclination); the mean Twin Sisters direction thus lies about 105° clockwise in declination and is some 14.5° shallower in inclination than the direction that should have prevailed at its present location at the time it was emplaced. The "expected" direction based on a Cretaceous magnetization time lies even further to the west. It seems unarguable, then, that the Twin Sisters Intrusion has experienced some tectonic disturbance since it became magnetized.

Geological evidence gathered by Misch and alluded to by Christensen [3] suggests that the Twin Sisters Intrusion has been tilted by post-emplacement folding of the Shuksan thrust. From fig. 19 of Christensen and an examination of regional structural trends (to which the long-dimension of the intrusion is substantially parallel), it seems possible that the intrusion may have been rotated a maximum of about 40° about an axis trending roughly N20W. Such a rotation would carry the "expected" Early Tertiary direction to the middle of the first quadrant of the stereographic projection, reducing but not eliminating the discrepancy between measured and predicted directions. In fact, the only way the discrepancy can be accounted for by simple tilt is to invoke very substantial folding about some northeasterly axis, as a high angle to other structural elements in the area. Such an explanation for the discrepant Twin Sisters magnetic direction seems improbable; more likely some degree of clockwise rotation about a vertical axis also is required.

If the Twin Sisters Intrusion were an isolated instance of a discordant pole position it would be easy to dismiss, especially in view of its complicated structural setting. In fact, however, it is merely one of many such poles that have been determined for rock units in the westernmost Cordillera over the past two decades. Elsewhere [9] I have compiled a list of 17 such discordant poles (see Fig. 3), using the "A-group" data of Hicken et al. [7] supplemented by newer poles that met the same criteria of reliability. Nearly all of these poles show clockwise rotation relative to their "expected" directions, and a majority also show a significant flattening of inclination. The list includes rock units ranging in age from Triassic to Pliocene, with present locations scattered from Alaska to Baja California. The work resulting in these discordant pole positions was done over the last 17 years by 19 investigators working in at least 10 separate laboratories. Coincidence and experimental difficulties thus seem ruled out; a physical explanation common to the entire western edge of the North American plate and acting continually or intermittently throughout the Mesozoic and Tertiary seems required, or at least strongly suggested, by the consistency of the data. In turn, the fact that nearly all of the discordant poles show clockwise rotation of declination and/or flattening of inclination strongly suggests distributed right-lateral shear as the dominant distribute mechanism. As will be shown more fully in a later paper, a zone of distributed right-lateral simple shear at the edge of the continent might displace rigid crustal blocks within the zone by northwestward translation, clockwise rotation, or both. Thus either inclination or declination or both could be affected. Parts of the western margin of the North American plate are in shear at the present time, and the plate-tectonic analyses of Atwater [10] and Stone and Packer [11] indicate that this has been the case, at least intermittently, for some time. The geological record also seems to support some degree of microcontinent-tectonics, involving very large northward displacements, within the western Cordillera (e.g. [12,13]). It would appear that with the additional data from paleomagnetism, enough evidence has accumulated to warrant consideration of microcontinent-tectonics driven by right-lateral shear in future tectonic syntheses. These ideas are being further developed for publication elsewhere.

Acknowledgments

So many students and colleagues have helped me with this project that it is impossible to name them all. Special thanks are owed to S. Beske-Diehl, R. Cockerham, J. Diehl, J. Duemmel, J. Gaston, H. Johnson, R. Keller, H. Kinoshita, K. Othberg, and J. Whitney, for direct assistance in the field and laboratory. This work was supported by NSF project GA-31895.

References

1 D.M. Ragan, The Twin Sisters Dunite, Washington, in: Ultramafic and Related Rocks, ed. P.J. Wyllie (John Wiley and Sons, New York, 1967) 160.

2 G.A. Thompson, Aeromagnetic and Bouguer Gravity map of Twin Sisters Dunite. Northwestern Washington, U.S. Geol. Survey GP Map 901 (1973).
3 N.I. Christensen, Fabric, seismic anisotropy, and tectonic history of the Twin Sisters Dunite, Washington, Geol. Soc. Am. Bull. 82 (1971) 1681.
4 P. Misch, Tectonic evolution of the Northern Cascades of Washington State, in: Tectonic History and Mineral Deposits of the Western Cordillera, Can. Inst. Min. Metall., Spec. Vol. 8 (1966) 101.
5 H. Kinoshita and M.E. Beck, Jr., Preliminary report of the paleomagnetism of the Twin Sisters Dunite Intrusion, Washington, U.S.A., J. Geomagnet. Geoelectr. 23 (1971) 401.
6 M.W. McElhinny, Paleomagnetism and Plate Tectonics (Cambridge, 1973) 357 pp.
7 A. Hicken, E. Irving, L.K. Law and J. Hastie, Catalogue of paleomagnetic directions and poles, Publ. Earth Physics Br. 45, No. 1 (Ottawa, 1972) 135.
8 E. Irving and J.K. Park, Hairpins and superintervals, Can. J. Earth Sci. 9 (1972) 1318.
9 M.E. Beck, Jr., Paleomagnetism and tectonics of the western Cordillera of North America, in: Proceedings of the Takesi Nagata Conference, ed. M. Fuller (Pittsburg, 1975) in press.
10 T. Atwater, Implications of plate tectonics for the Cenozoic evolution of western North America, Geol. Soc. Am. Bull. 81 (1970) 3513.
11 D. Stone and D.R. Packer, Plate motions, hot spots, paleomagnetism and the tectonics of western North America and Alaska, Geol. Soc. Am. Bull. (1975) in press.
12 J.W.H. Monger and C.A. Ross, Distribution of Fusulinaceans in the western Canadian Cordillera, Can. J. Earth Sci. 8 (1971) 259.
13 D.L. Jones, W.P. Irwin and A.T. Ovenshine, Southeastern Alaska – a displaced continental fragment?, U.S. Geol. Survey Prof. Paper 800-B (1972) B211.